Conceptual Data Model

- Entities:
- Customer
- Order
- OrderSummary
- CustomerAggregateSpend

- Relationships:
- A Customer may have multiple Orders.
- Orders are linked to a Customer via CustId.
- OrderSummary is derived from joining Customer and Order.
- CustomerAggregateSpend aggregates spend per customer per date from OrderSummary.

- Entity Definitions:
- Customer: Represents individuals or entities placing orders.
- Order: Represents each transaction or purchase.
- OrderSummary: Historical summary of orders with customer and order details (SCD Type 2).
- CustomerAggregateSpend: Aggregated spending amounts for each customer by date.

Logical Data Model

- Entities and Attributes:
- Customer
- CustId (PK)
- Name
- EmailId
- Region
- Order
- OrderId (PK)
- ItemName
- PricePerUnit
- Qty
- Date
- CustId (FK → Customer.CustId)
- OrderSummary
- CustId
- Name

- EmailId
- Region
- OrderId
- ItemName
- PricePerUnit
- Qty
- Date
- StartDate (SCD2)
- EndDate (SCD2)
- ActiveFlag (SCD2 status)
- (PK: CustId, OrderId, StartDate)
- CustomerAggregateSpend
- Name
- TotalAmount
- Date
- (PK: Name, Date)

- Keys & Cardinality:
- One-to-many from Customer to Order (CustId).
- OrderSummary maintains historical variants (SCD2 logic) per (CustId, OrderId).
- CustomerAggregateSpend groups by customer Name and Date.

- Normalization Notes:
- 1NF: All attributes hold atomic values.
- 2NF: All non-key fields depend on whole key.
- 3NF: No transitive dependencies among non-keys.

- Explicit Business Rules:
- CustId must be unique in Customer.
- EmailId must be valid and unique.
- Orders require matching CustId in Customer.
- Null and duplicate records are not allowed.

Physical Data Model

- Tables and Structures:
- customer
- CustId: INTEGER/UUID, NOT NULL, PRIMARY KEY

- Name: VARCHAR(100), NOT NULL
- EmailId: VARCHAR(255), NOT NULL, UNIQUE
- Region: VARCHAR(50),
- order
- OrderId: INTEGER/UUID, NOT NULL, PRIMARY KEY
- ItemName: VARCHAR(100), NOT NULL
- PricePerUnit: DECIMAL(10,2), NOT NULL
- Qty: INTEGER, NOT NULL
- Date: DATE, NOT NULL
- CustId: INTEGER/UUID, NOT NULL, FOREIGN KEY → customer.CustId
- ordersummary
- CustId: INTEGER/UUID, NOT NULL
- Name: VARCHAR(100), NOT NULL
- EmailId: VARCHAR(255), NOT NULL
- Region: VARCHAR(50),
- OrderId: INTEGER/UUID, NOT NULL
- ItemName: VARCHAR(100), NOT NULL
- PricePerUnit: DECIMAL(10,2), NOT NULL
- Qty: INTEGER, NOT NULL
- Date: DATE, NOT NULL
- StartDate: DATE, NOT NULL
- EndDate: DATE, NULL
- ActiveFlag: CHAR(1), NOT NULL ('Y'/'N')
- PRIMARY KEY (CustId, OrderId, StartDate)
- Indexes on ActiveFlag, Date
- Partitioning: By Date or Region ()
- customeraggregatespend
- Name: VARCHAR(100), NOT NULL
- TotalAmount: DECIMAL(18,2), NOT NULL
- Date: DATE, NOT NULL
- PRIMARY KEY (Name, Date)
- Indexes on TotalAmount
- Partitioning: By Date ()

- Constraints
- PKs, FKs, and unique constraints as above.

- Check: PricePerUnit > 0, Qty > 0
- Not Null constraints on all mandatory fields.

- Performance Optimization Notes
- Indexes on search/join columns (CustId, OrderId, Date, ActiveFlag).
- Cluster/partition large tables by frequently filtered columns.
- Maintain SCD logic in ordersummary for historical tracking.

Naming Conventions

- Table names: all lowercase, singular noun (e.g., customer)
- Column names: camelCase or underscores (e.g., CustId, PricePerUnit)
- PK/FK conventions: PK as TableId, FK as referenced PK name

Data Dictionary

- CustId: Customer Identifier, Unique, PRIMARY KEY (customer)
- Name: Customer Name, VARCHAR(100)
- EmailId: Customer Email, VARCHAR(255)
- Region: Customer Region, VARCHAR(50)
- OrderId: Order Identifier, Unique, PRIMARY KEY (order)
- ItemName: Product or Service Name, VARCHAR(100)
- PricePerUnit: Price per single unit, DECIMAL(10,2)
- Qty: Quantity, INTEGER
- Date: Transaction or aggregation date, DATE
- TotalAmount: Computed field (PricePerUnit * Qty), DECIMAL(18,2)
- StartDate: Record effective from, DATE
- EndDate: Record effective until, DATE (nullable)
- ActiveFlag: SCD2 status, 'Y' or 'N'
- (Other attributes if source expands)

Text-Based ERD (Entity-Relationship Description)

- Customer (1) ——< (M) Order
- Customer.CustId = Order.CustId (FK)
- OrderSummary is a denormalized SCD2 table combining Customer + Order, with effective dating
- CustomerAggregateSpend is derived from OrderSummary, grouping by Name and Date for total spending

: Data types, additional constraints, partitioning, and clustering to be determined by implementation requirements.